AGENTIC AI ASSISTANCE FOR CONTENT CREATORS

Your An End-to-End AI-Driven Content Automation System for Research, Video Generation, and Multi-Platform Publishing

N.BHARGAVA TEJA - ME22B2006
Supervisor : Dr. M Arul Kumar

Problem Statement

Content creation is a time-consuming multi-stage process involving:

- manual research,
- script preparation,
- repeated video recording and retakes,
- voice-over generation,
- editing intros, outros, and backgrounds,
- multi-platform publishing,
- and workflow monitoring.

For busy professionals, these repetitive tasks create productivity loss, inconsistency, and cognitive overload.

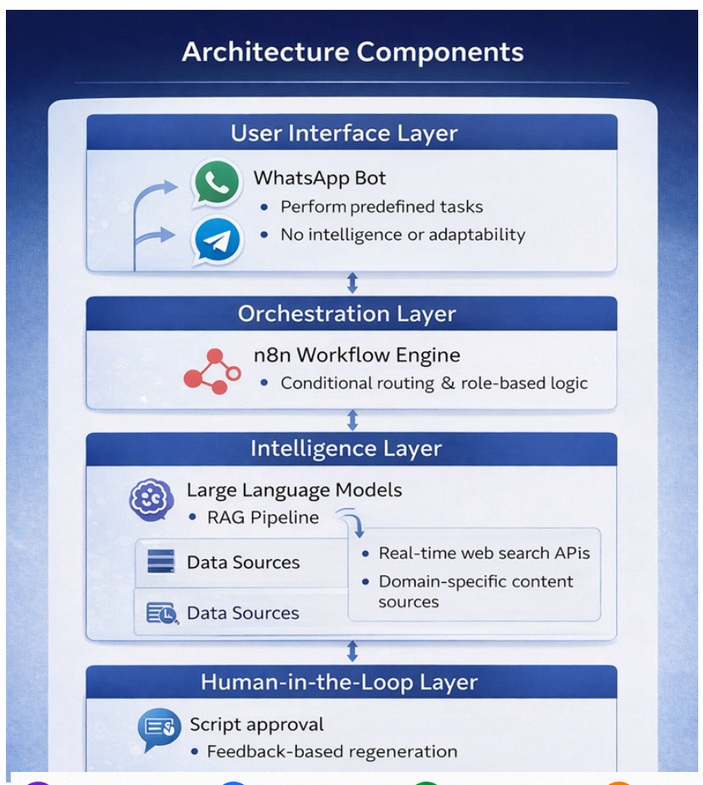
Objectives

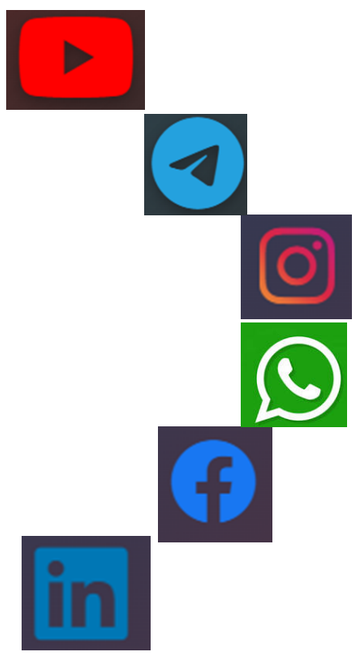
- Develop a complete Agentic AI automation pipeline
- Implement RAG-based grounded script generation
- Automate AI-based video generation and editing
- Integrate avatar cloning and voice cloning
- Enable multi-platform publishing automation
- Develop centralized logging and monitoring systems

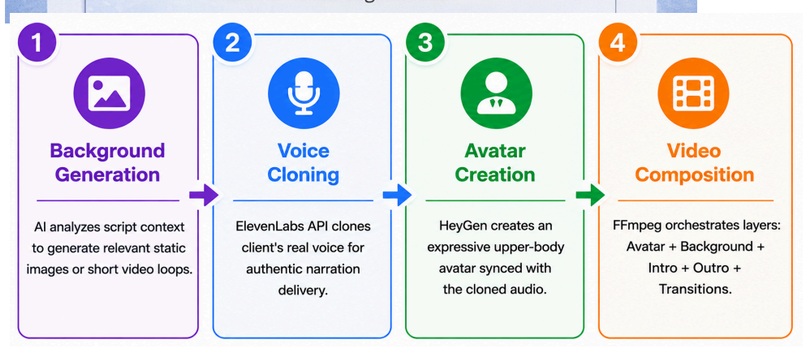
Assumptions

- Users have access to WhatsApp or Telegram
- API services remain accessible during execution
- Retrieved content sources are reliable and relevant
- Human approval is mandatory before publishing

Concept Design







Bill of Materials

COMPONENTS	COST
n8n, ffmpeg, Ollama LLMS	Open-source (Self hosting)
Serp, Tavily, Whatsapp, Telegram APIS	Completely Free or free credits limit usage
Heygen API	$ 0.00167 /sec of video
ElevenLabs API	$ 0.001/ sec of audio
TOTAL	$ 0.25/minute of video (Approx)

Prototype

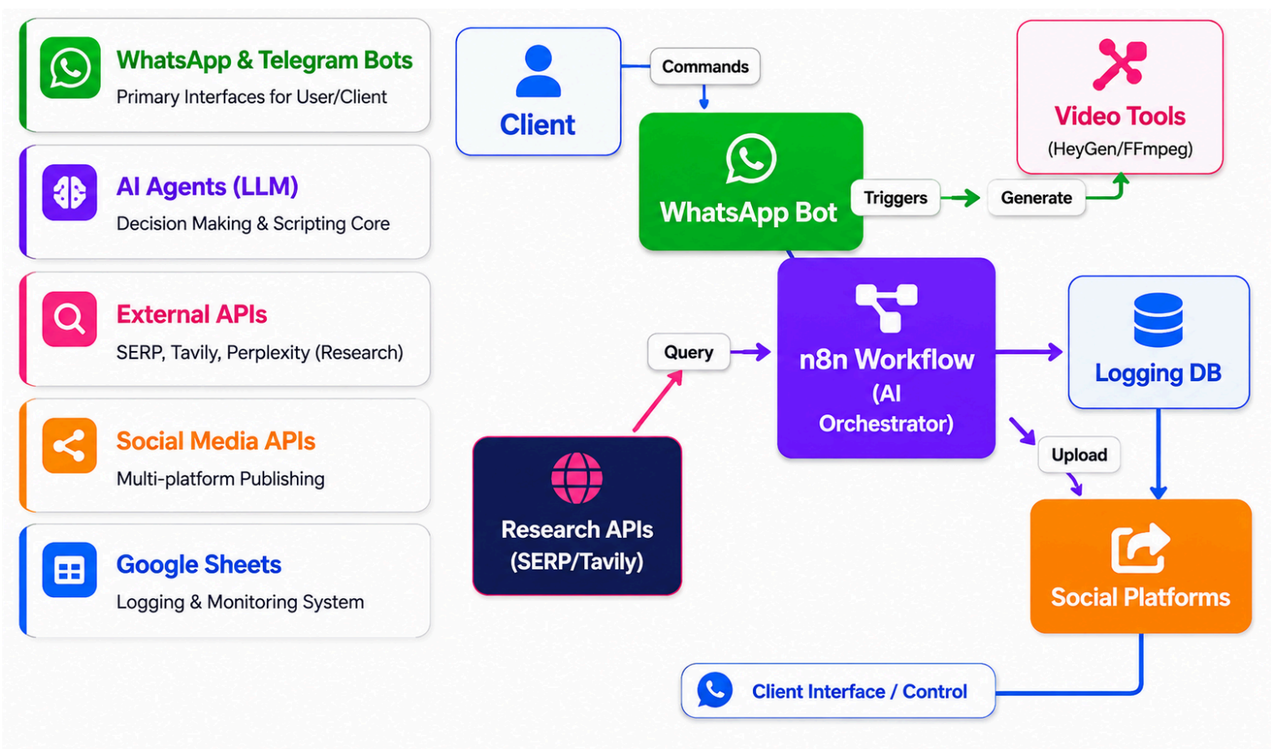
- WhatsApp and Telegram bots developed for workflow triggering and user interaction
- RAG-based AI pipeline implemented for real-time research and script generation
- Automated video generation integrated using HeyGen, ElevenLabs, and FFmpeg
- Multi-platform publishing, logging, and real-time error alert systems successfully tested

Outcome & Inferences



References

- P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” NeurIPS, 2020.
- n8n Documentation – https://n8n.io



COMPLETE ARCHITECTURE

Indian Institute of Information Technology, Design & Manufacturing
IIITD&M Kancheepuram, Chennai - 600 127
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